Kashaf Fatima

AI Engineer | Agent Evaluation, Governed Agent Runtimes, Deterministic Verification
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github.com/kashaffatimajaffrey-design | kashaffatimajaffrey-design.github.io/Portfolio

SUMMARY

AI engineer who builds the checks that decide whether an agent is ready to ship, and keeps enforcing them after it does: eval sets built from real user queries, known-answer tests reported with confidence bounds, deterministic gates and human approval before irreversible actions, and data access enforced in the database rather than the model.

EXPERIENCE

Occuris AI, Founder & Lead AI Engineer | Karachi (hybrid), team of three | Jan 2026 – Present

- Caught 87 of 100 planted supply problems (95% CI 79–92%) across 300+ items in a six-month pilot with NKE Technica (US industrial equipment), by evaluating Occuris Command's procurement and inventory agents against a known-answer set and setting thresholds to prefer false alarms over misses.
- Put a human approval gate in front of every irreversible action by agents that run unprompted on schedules and events (LangGraph, FastAPI, React), by routing those actions through Occuris Echo for sign-off over Gmail or Slack.
- Kept the platform free of single-provider lock-in, with every answer traceable to the model that produced it, by running agents interchangeably on Claude, Grok or Ollama and recording the model on each response.
- Made tenant isolation a database guarantee rather than an app-code promise, by deriving the tenant from the verified Supabase Auth session and enforcing row-level security on every table.

ProForecast, AI/ML Engineer | UK company (remote) | Sep 2024 – Mar 2026

- Raised PROPILOT's answer accuracy from 43% to 82%, measured on an eval set I built from real Sage 50 client queries and scored on intent and the figure returned, by fixing semantic-search and intent failures with BERT/RoBERTa embeddings.
- Delivered 12-month forecasts and PROPILOT's answer tables for clients in Germany, the US and elsewhere, by building RAPIER's pipeline over 18 months of sensitive client finance and warehouse data on HDFS, Spark and Snowflake.
- Shipped every change to both production products (Django, DRF, TensorFlow) through pull-request review by the engineering lead, and wrote both products' technical documentation.

10Pearls, Data Science Intern (Shine internship) | alongside Occuris | Jul 2026 – Sep 2026

- Cut Karachi air-quality forecast error (RMSE) from 8.63 to 7.74 (R^2 0.21 to 0.36) by adding known-future weather to Pearls AQI Predictor, a three-day forecaster that hourly GitHub Actions pipelines retrain and promote from seven model families with no manual step (scikit-learn, SHAP, Streamlit).
  github.com/kashaffatimajaffrey-design/pearls-aqi-predictor | live dashboard

PROJECTS

Occuris Echo, governed voice agent | Python, Claude, Gmail, Calendar, Slack, Sheets

- Reached 57/57 known-answer cases with 0 silent failures, against 3 silent failures with the deterministic layer switched off, by settling requests with deterministic rules before Claude, asking instead of guessing, requiring confirmation before any irreversible action, and blocking duplicate sends.
  github.com/kashaffatimajaffrey-design/occuris-echo | www.loom.com/share/6bd1cf3edd624f7abde12051eb732f79

supabase-ai-eval, eval-first RAG agent | Python, FastMCP, pgvector, Claude, React

- Made answer quality measurable on every run, through LLM-as-judge scores logged to Postgres and a live dashboard, by building hybrid HNSW and full-text retrieval served to other agents as MCP tools.
  github.com/kashaffatimajaffrey-design/supabase-ai-eval | supabase-ai-eval.vercel.app

Supabase Lite Conformance Kit, differential testing | TypeScript, supabase-js, PostgreSQL, SQLite, GitHub Actions

- Traced 48 failing cases to 22 root causes, ranked with silent wrong data first, by running 81 supabase-js calls against Postgres and SQLite, diffing the results, and citing the PostgREST and GoTrue source that defines correct behaviour.
- Cut silent wrong-data findings from 15 to 12 and raised passing cases from 33 to 36 with no other case regressing, by fixing two root causes upstream with 13 new tests and a CI report republished on every push.
  kashaffatimajaffrey-design.github.io/supalite-conformance | github.com/olirice/supabase-lite/pull/1

PUBLICATION

A Known-Answer Audit for Deployed Measurement Systems: Procedure and a Case Study. Sole author, Zenodo preprint, 2026. Found two components silently returning fallback constants for about 70% of observations by planting ground truth in my own pipeline, and withdrew the headline claim when bootstrap confidence intervals did not support it. doi.org/10.5281/zenodo.22395202

SKILLS

Evaluation & statistics: known-answer and differential testing, LLM-as-judge, regression checks in CI, bootstrap and binomial confidence intervals, error-rate estimation from samples

Agents & LLMs: LangGraph, MCP, RAG, pgvector, human-in-the-loop approval, Claude, Grok, OpenAI, Gemini, Ollama, BERT, PyTorch, TensorFlow

Engineering: Python, TypeScript, FastAPI, Django, React, PostgreSQL, Supabase RLS, Rust, Spark, Snowflake, AWS, pytest, Vitest, Playwright, GitHub Actions

EDUCATION

Bahria University Karachi, BS Computer Science (AI/ML specialization) | Expected January 2027